Project 4:-Maven Selenium Project – Automatic Login

Website used for testing:

https://www.saucedemo.com/

Step 1: Create a Maven Project:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven
-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/test/MyMavenSeleniumApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.913 s
[INFO] Finished at: 2026-03-17T20:56:48+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2: Edit pom.xml, add selenium dependency:

Open the **pom.xml** file and add the Selenium dependency.

<dependency>

<groupId>org.seleniumhq.selenium</groupId>

<artifactId>selenium-java</artifactId>

```
<version>4.29.0</version>
</dependency>
```

Step 3: Configure Maven Build Plugins

Add the following plugins in the **pom.xml** file.

```
<build>
<plugins>

<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>

<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>

</plugin>

<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.0.1</version>

<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
```

```
</plugins>
</build>
```

Step 5: Inspect required elements in the Website

Before writing Selenium code, we must inspect the HTML elements.

Step 6: Create App.java

Java Code:

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class App
{
    public static void main(String[] args) throws InterruptedException
    {
        WebDriver driver = new ChromeDriver();

        Thread.sleep(2000);
        driver.get("https://www.saucedemo.com/");

        Thread.sleep(2000);
        driver.manage().window().maximize();
```

```
        Thread.sleep(2000);

        driver.findElement(By.id("user-name")).sendKeys("standard_user");


        Thread.sleep(2000);

        driver.findElement(By.id("password")).sendKeys("secret_sauce");


        Thread.sleep(2000);

        driver.findElement(By.id("login-button")).click();

    }}
```

Step 7: Build the Maven Project

Run the following command:

mvn clean install

This command compiles the project and builds the JAR file.

```
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.733 s
[INFO] Finished at: 2026-03-17T21:08:14+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$
```

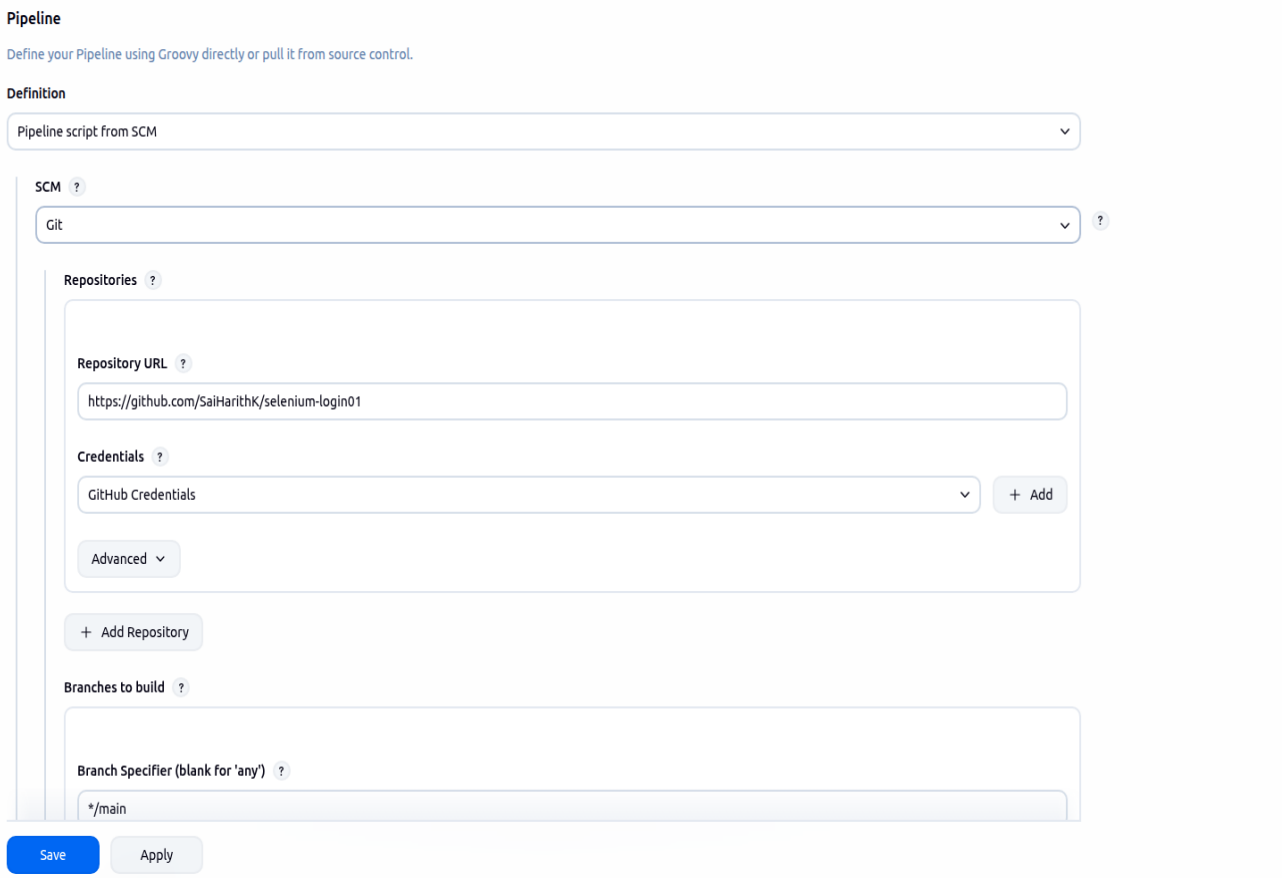
mvn exec:java -Dexec.mainClass=”com.example.App”

```
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenSeleniumApp01 ---
```

Step 8: Pipeline

-login to jenkins account -> Daashboard -> new item

-create a new item and configure the following settings



Step 9: Edit Jenkins file

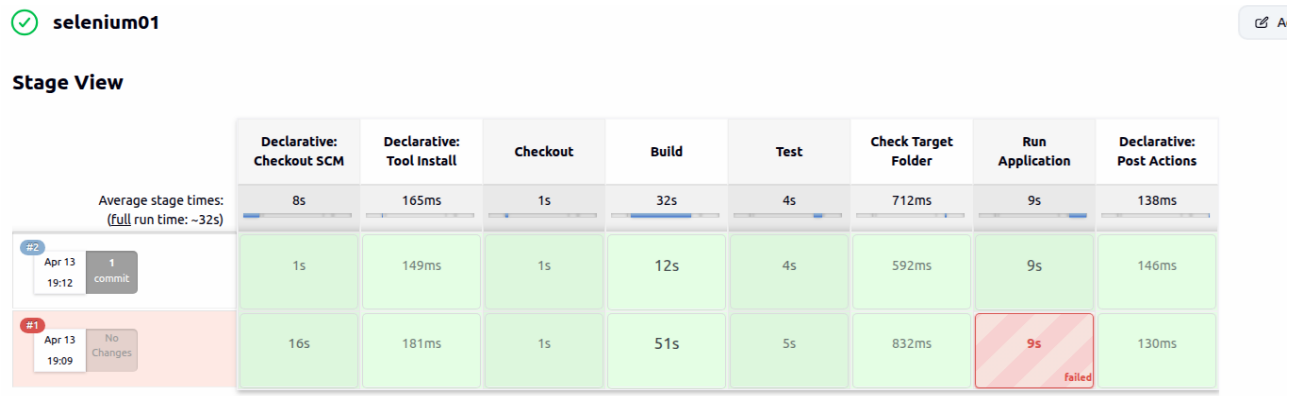
-Create a Jenkinsfile and add the following code:

```
pipeline {
    agent any  // Use any available agent


    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/SaiHarithK/selenium-login01.git'
            }
        }
```

```
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}
       stage('Run Application') {
    steps {
        sh 'java -jar target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar'
    }
}
}
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 10: Build

Click on build now and check if its successfull